Least squares regression line

1 Given an example of a correlation coefficient value that suggests a strong linear relationship between two variables.

2 Find the equation of the least squares regression line if:

a An x-value of 5 gives a predicted value of $y = 9$, and an x-value of 8 gives a predicted value of $y = 3$.

b An x-value of 4 gives a predicted value of $y = 3$, and an x-value of 6 gives a predicted value of $y = 7$.

3 Lachlan analyzes data comparing average outdoor temperature with the cost of air conditioning for his ice cream store. He fits the data to a linear model and finds the correlation coefficient to be 0.47.
Based on the analysis Lachlan made, describe the association between the increase in outdoor temperature and air conditioning costs.

4 For each of the following sets of data:

i Use technology to calculate the correlation coefficient to two decimal places.

ii Describe the statistical relationship between the two variables.

iii Use technology to form an equation for the least squares regression line. Round all values to one decimal place.

a

x	15.7	13.1	16.1	11	18.6	15.8	12.7	12.8	16.8	14.3
y	28.3	28.8	28.4	29	27.9	28.4	28.5	29	28.5	28.6

b

x	48.50	38.50	39.70	42.20	29.60	42.60	23.80	47.60	20.80
y	166.10	63.40	143.90	142.20	148.70	174.00	-52.30	195.50	65.50

5 A cafe manager collected data about sales [illegible]t cocoa during a winter weekend. The data included the outside temperature and the number of hot cocoa sold every two hours.

Outside temperature in $^\circ$F, t	0	2	5	2	1	3	6	4
No. hot cocoa sold, n	20	16	3	16	18	17	8	12

a Write the linear regression equation for this data set. Round all values to the nearest hundredth.

b State the correlation coefficient for this data set, to the nearest hundredth.

c What does the sign of the correlation coefficient suggest in the context of this problem?

6 The forecast maximum temperature, in degrees Celsius, and the observed maximum temperature are recorded to determine the accuracy in the temperature prediction models used by the weather bureau.

a Calculate the correlation coefficient for these temperatures, rounding your answer to two decimal places.

b Describe the statistical relationship between these two variables.

c Use technology to find the equation for the least squares regression line of y on x.

d Use your regression line to predict the observed maximum temperature on a day in the same month when the forecast was 25°C.

e Which of the following forecast temperatures would give the most reliable predicted temperature? Explain your answer.

$$39^\circ\text{C}, 25^\circ\text{C}, \text{ or } 32^\circ\text{C}$$

Forecast (x)	Observed (y)
30.10	31.20
27.20	27.80
26.90	24.30
29.00	27.60
30.20	31.60
34.20	31.50
33.20	33.70
36.90	34.30
27.10	25.00
30.10	31.90

7 Research on the number of cigarettes sm[illegible] during pregnancy and the birth weights of the newborn babies was conducted and results displayed in the table below.

a Calculate the correlation coefficient for the data, rounding your answer to three decimal places.

b Describe the statistical relationship between these two variables.

c Use technology to find the equation of the least squares regression line of y on x. Round all values to two decimal places where necesssary.

d Use your regression line to predict the birth weight of a newborn whose mother smoked on average 5 cigarettes per day.

e Comment on the reliability of your prediction.

Average number of cigarettes per day (x)	Birth weight in kilograms (y)
46.30	3.90
13.00	5.80
21.40	5.00
25.00	4.80
8.60	5.50
36.50	4.50
1.00	7.00
17.90	5.10
10.60	5.50
13.40	5.10
37.30	3.80
18.50	5.70

8 A team of salespeople submit their expenses and their sales for the month of March to their manager.

a Calculate the correlation coefficient for the data, rounding your answer to two decimal places.

b Describe the statistical relationship between these two variables.

c Use technology to find an equation for the least squares regression line of y on x. Round all values to two decimal places.

d Use your regression line to predict the expenses of a person in this department who made sales of \$6000 for the month. Round your answer to two decimal places.

e Comment on the reliability of your prediction.

Sales (hundreds of dollars) (x)	Expenses (dollars) (y)
55.7	43.8
48	28.9
62.7	73.9
23.4	11.9
23.8	9
60.6	50
20.2	20.2
52.1	28.2
23.4	19.2
33.3	31.1

9 During an alcohol education programme, [redacted]ults were offered up to 6 drinks and were then given a simulated driving test where they scored a result out of a possible 100. The results are displayed in the following table:

Number of drinks (x)	3	2	6	4	4	1	6	3	4	2
Driving score (y)	66	61	43	58	56	73	31	64	55	62

a Calculate the correlation coefficient for the data, rounding your answer to two decimal places.

b Describe the statistical relationship between the two variables.

c Use technology to find the equation of the least squares regression line of y on x. Round all values to one decimal place.

d Use your regression line to predict the driving score of a young adult who consumed 5 drinks. Round your answer to one decimal place.

e Comment on the validity of your prediction.

10 Research has been conducted to deduce whether there is any relationship between the number of pirated downloads as a percentage of total downloads, x, and the incidences of malware infection in computers per 1000 computers, y.

x	40.5	61.1	77.4	34.6	65.3	63.8	56	66.5	34.3	20.1	68.7	50.8
y	3.7	7.1	16.7	10.6	5.2	15.8	12.7	12.8	1.5	0.9	8.5	12.9

a Calculate the correlation coefficient between the two variables, rounding your answer to two decimal places.

b Describe the statistical relationship between these two variables.

c Use technology to form an equation for the least squares regression line of y on x. Round all values to two decimal places.

d Use your regression line to predict the number of computers in 1000 that you would expect to be affected by malware if in that region 80% of downloads were pirated.

e Describe the reliability of this prediction.

11 A sample of families were interviewed ab[redacted]eir annual family income, x, and their average monthly expenditure, y. Results are displayed in the table below:

Annual income (x)	Average monthly expenditure (y)
66 000	1100
75 000	1700
65 000	1400
73 000	1300
54 000	600
90 000	1800
87 000	1100
87 000	1500
94 000	1800
96 000	2200

a Calculate the correlation coefficient between the two variables, rounding your answer to two decimal places.

b Describe the statistical relationship between the two variables.

c Use technology to find the equation for the least squares regression line of y on x. Round all values to three decimal places.

d Use your regression line to predict the monthly expenditure for a family whose annual income is $\$80\,000$.

e Which of the following annual incomes would give the more reliable prediction of average monthly expenditure? Explain your answer.

$\$99\,000$, $\$51\,000$ or $\$80\,000$

12 The results (as percentages) for a practice spelling test and the real spelling test were collected for 8 students:

Practice (x)	56.30	79.00	59.40	77.00	71.60	64.40	61.60	68.20
Real (y)	48.90	69.30	51.90	66.00	62.50	56.10	54.50	59.20

a Calculate the correlation coefficient for the scores, rounding your answer to three decimal places.

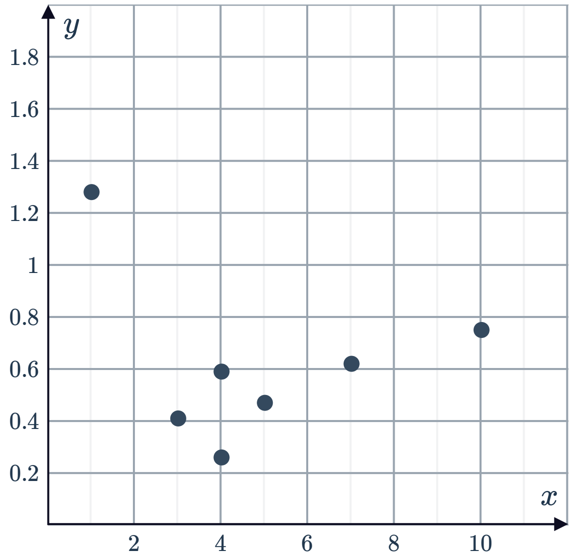
b Describe the statistical relationship between these two variables.

c Use technology to find the equation for the least squares regression line of y on x. Round all values to two decimal places.

d Use your regression line to predict the real spelling test result of a student who scored 60% in their practice spelling test. Round the answer to two decimal places.

e Comment on the validity of this prediction.

13 The data in the table and scatter plot belo[illegible]w the frequencies per month of online marketing emails sent out to subscribers, compared with the proportion of subscribers who click and open the email.



Frequency (x)	Proportion click and open (y)
3	0.41
4	0.59
1	1.28
5	0.47
4	0.26
7	0.62
10	0.75

a Calculate the correlation coefficient between these two variables, rounding your answer to two decimal places.

b Which piece of data appears to be an outlier?

c Remove the outlier and recalculate the correlation coefficient.

d Describe the statistical relationship between the two variables.

e Use technology to find the equation for the least squares regression line of y on x (with the outlier removed). Round values to two decimal places.

f Use your regression line to predict the proportion of emails opened if they're sent 20 times a month.

g Comment on the validity of your prediction.

Calculating residuals

14 Complete the following tables of residuals:

a

x	y	$\hat{y}$	Residual
10	29	25	
14		37	-8
7	20		4
6		13	8
20	58	55	
13	26		-8

b

x	y	$\hat{y}$	Residual
12.8		-33.3	0.6
11.4	-30.3	-30.3	
11.1		-29.7	0.2
6.5	-21.5		-1.4
10	-28.5	-27.4	-1.1

15 The following tables show sets of data $(x,$ [redacted]d the predicted $\hat{y}$ values based on a least squares regression line. Complete the tables by finding the residuals:

a

x	1	3	5	7	9
y	22.7	22.3	24.2	21.8	21.5
$\hat{y}$	25.2	23.4	21.6	19.8	18
Residuals					

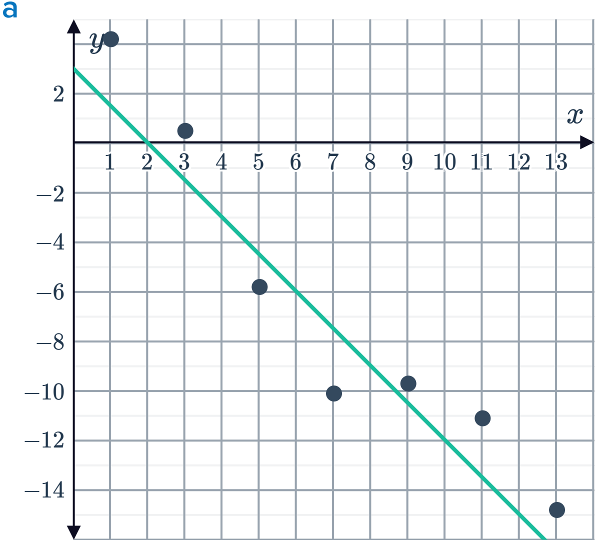
b

x	5	6	7	8	9
y	37.7	37.2	21.1	27.1	44
$\hat{y}$	28.9	30.4	31.9	33.4	34.9
Residuals					

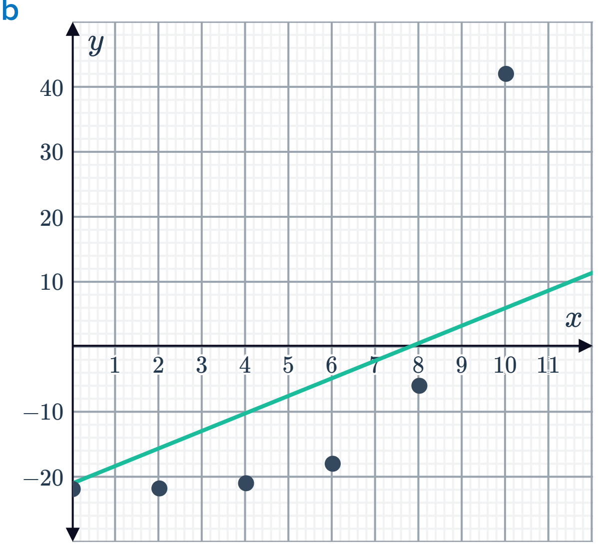
Residual plots

16 Construct the residual scatter plot of the following data sets:

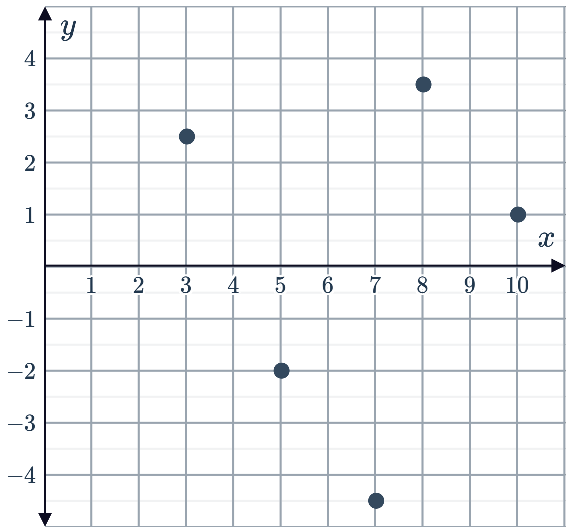
a



b



17 The residual plot for a set of data is shown. Draw the scatter plot showing the original data, if the linear regression line is given by $y = x + 6$.



18 For each of the data sets below:

- **i** Complete the table by finding the residuals.
- **ii** Plot the residuals on a scatter plot.
- **iii** Determine whether the original data has a linear or nonlinear relationship.
- **iv** Create a scatter plot for the original data.

a

x	y	$\hat{y}$	Residuals
2	14.5	12	
4	9	11	
6	12	10	
8	7	9	
10	10.5	8	

b

x	y	$\hat{y}$	Residuals
2	30	8	
4	20	10	
6	14	12	
8	12	14	
10	14	16	
12	20	18	

19 For each of the following data sets:

i Complete the table of residuals.

ii Plot the residuals on a scatter plot.

iii Determine if the model is a good fit for the data.

a The table shows a company's costs, C (in millions), in week W.
The equation $C = 3W + 4$ is being used to model the data.

W	C	Model value	Residual
1	7		
2	9		
5	21		
7	26		
10	34		
13	43		
15	48		
18	58		

b The table shows a company's revenue, R (in millions), in week W.
The equation $R = 3W + 5$ is being used to model the data.

W	R	Model value	Residual
2	7		
3	12		
5	19		
7	29		
9	37		
10	38		
12	41		
14	45		